



Machine Learning model For Predicting House Rent Prices in Ibadan: A data driven approach

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Background of the Study

- **Housing Rent in Nigeria:** Rent is a periodic payment by a tenant to a landlord for the occupation of a property (Onyejiaka et al., 2025). In Ibadan, rent prices vary widely due to location, infrastructure, and property type.
- **Limitations of Traditional Valuation:** Conventional methods rely on agent judgment and limited market data, leading to inconsistent, subjective, and time-consuming valuations.
- **Machine Learning in Real Estate:** ML algorithms, particularly regression-based models, have demonstrated high accuracy in housing price prediction tasks globally (Smith et al., 2024).
- **Linear Regression:** Models the relationship between a dependent variable (rent price) and independent variables (location, size, amenities, etc.), making it interpretable and computationally efficient.
- **Gap Identified:** No data-driven rent prediction tool exists specifically for the Ibadan housing market, motivating a localized ML solution.

Related Works

S/ N	Title / Authors / Year	Methodology	Major Results	Observed Weakness
1	Integrating Random Forest and XGBoost For Accurate House Rent Prediction — Singh et al. (2025)	Ensemble ML: Random Forest + XGBoost; regression on rental factors (location, sq ft, amenities)	High accuracy; combined model outperformed individual baselines; location and structural features as key predictors	High computational cost; potential overfitting; dataset-dependent
2	Real Estate Decision-Making: Price Prediction via Advanced ML Algorithms — Shbool et al. (2025)	SVR and Gradient Boosting Machine (GBM)	Reduced bias vs. traditional methods; high accuracy in valuation	Focused on Amman region; may need adaptation for other markets
3	Enhancing House Price Forecasting with Hybrid Data-Driven Feature Extraction — Soon et al. (2025)	Hybrid VMD-RF feature extraction + SVR/LSTM	High $R^2 = 0.97$; unified health index improves prediction	Relies on specific decomposition; complex hybrid setup
4	An Exploratory Approach to Composite Modelling for Real Estate Assessment — Hermans et al. (2023)	Composite modelling (exploratory AVM synthesis)	Improved accuracy via model combination	Exploratory; needs broader validation
5	A Survey of Methods and Input Data Types for House Price Prediction — Geerts et al. (2023)	Multiple ML regression: XGBoost, RF, SVR, MLP, MLR; hyperparameter tuning	XGBoost best performer; location, size, features identified as key	High computational cost; data quality sensitive
6	Transparent House Price Prediction Framework Using Ensemble Learning and ANOVA — Hussain et al. (2025)	Ensemble ML (XGBR, RFR, CBR); Genetic Algorithm tuning; SHAP/LIME explainability	Categorical Boosting + GA: $R^2 \approx 0.9973$; high interpretability	High complexity/training time; compute-heavy

Statement of Problem

Traditional housing price valuation methods are subjective, time-consuming, and inconsistent, relying heavily on human judgment and limited market data since there is no existing data-driven, ML-powered rent prediction system tailored specifically for the Ibadan housing market. Development of a data-driven rent prediction system using a Linear Regression model trained on historical Ibadan housing data, deployed as an interactive web application will be of help.

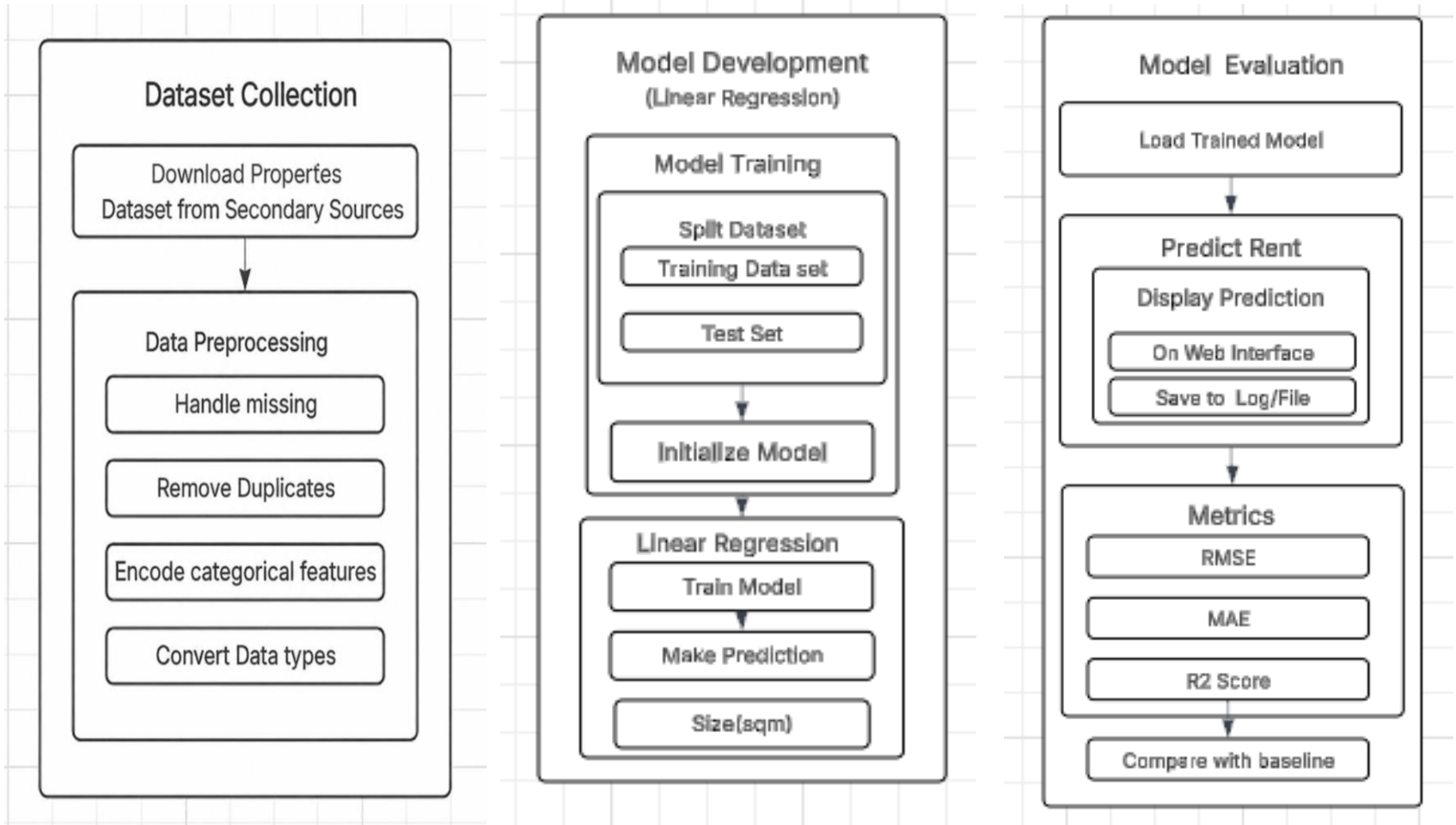
AIM AND OBJECTIVES

The aim is to develop a Linear Regression model for predicting house rent prices in Ibadan.

The objectives of this project are:

- Data preprocessing and feature engineering
- For predicting house rent using relevant housing features through machine learning model.
- To evaluate the performance and effectiveness of the developed linear regression model using appropriate statistical metrics

Methodology



Data collection and preprocessing

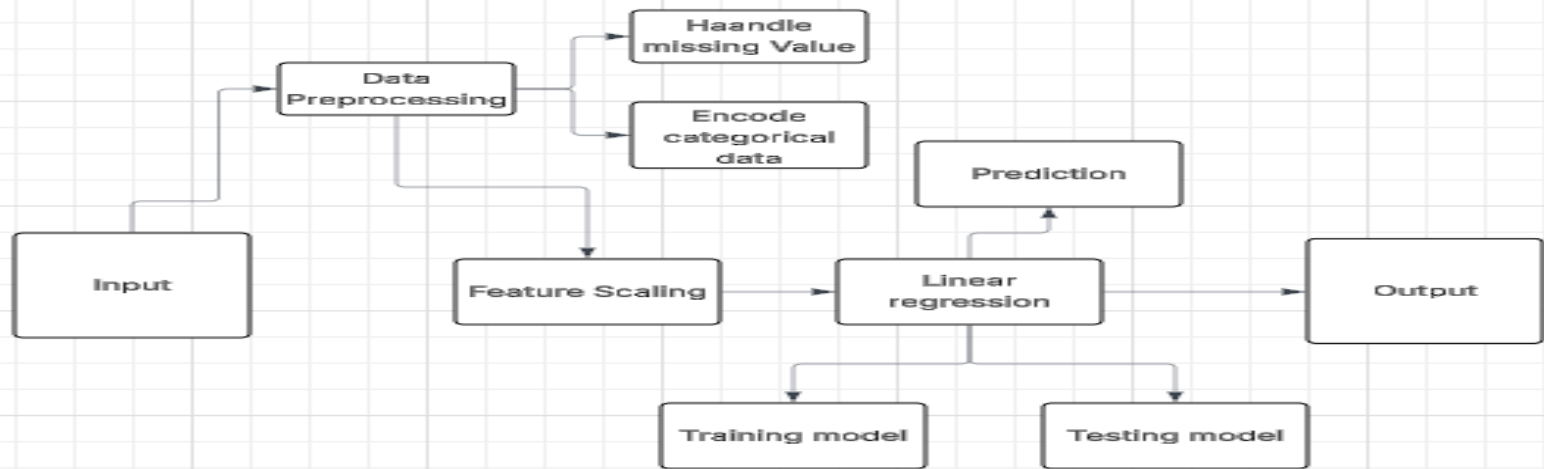
Model Development

Model Evaluation

Data Flow Diagram

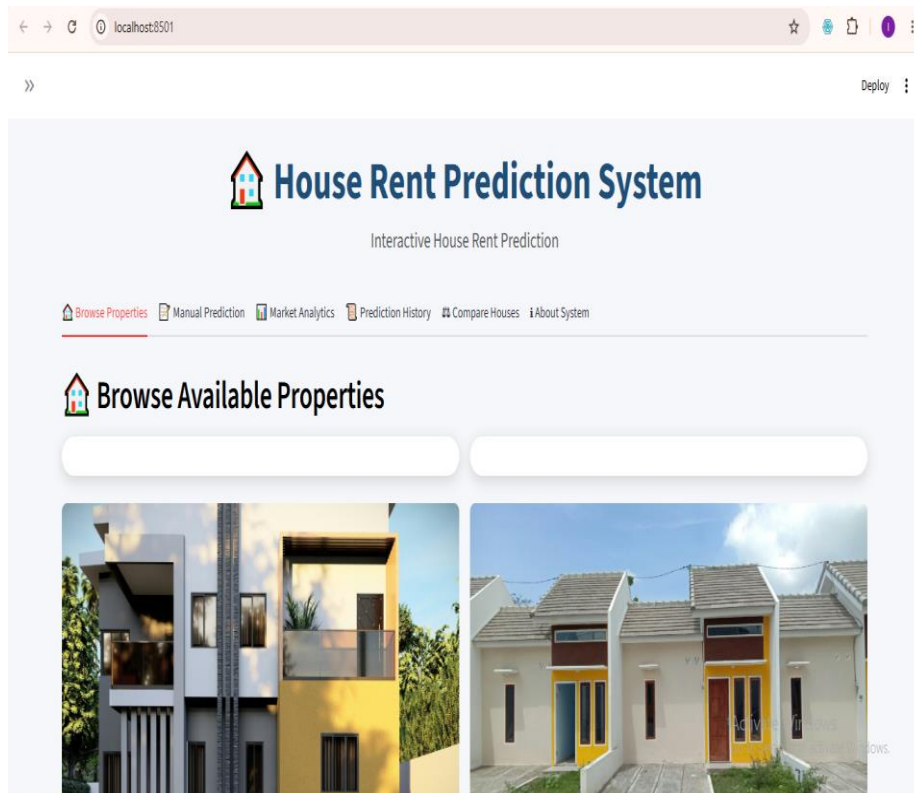


Level 0

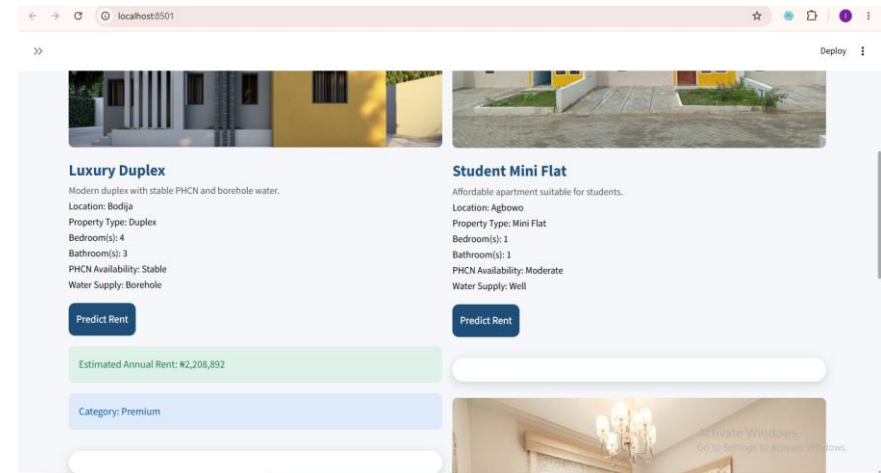


Level 1

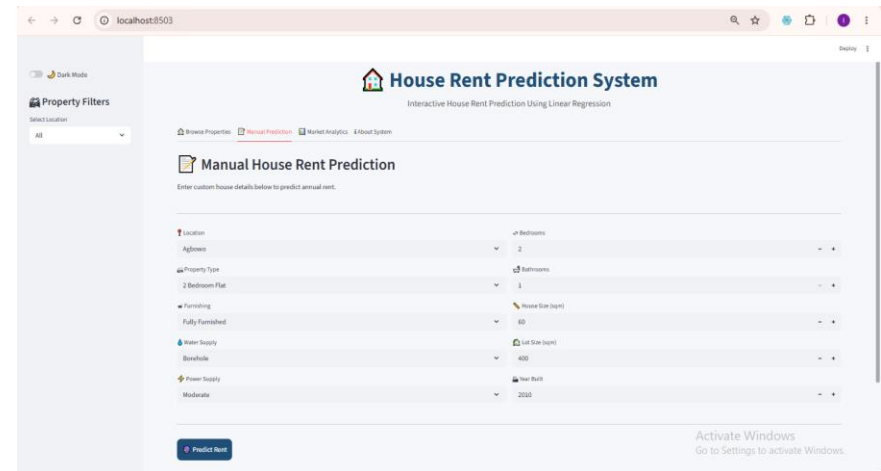
System Interface – Home & Manual Prediction



Home Interface (Browse Properties)

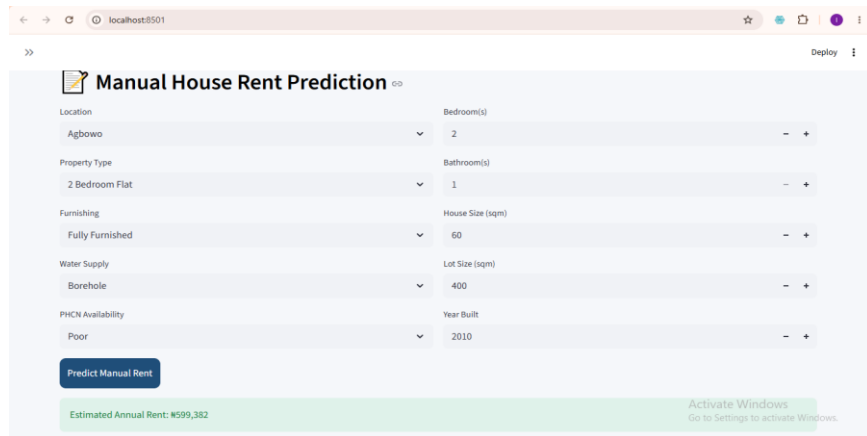


Property Gallery Interface



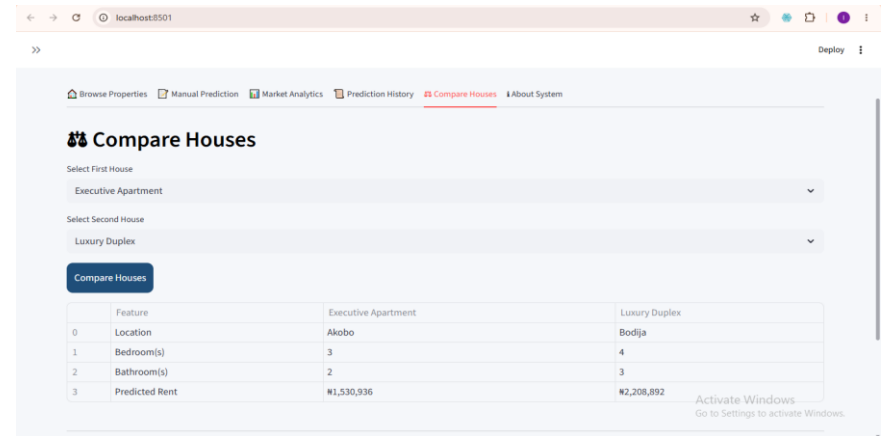
Manual Prediction Interface

System Interface – Property Gallery, Comparison & History



A web browser window showing the 'Manual House Rent Prediction' interface. It features a form with dropdown menus for Location (Agbowo), Bedroom(s) (2), Property Type (2 Bedroom Flat), Bathroom(s) (1), Furnishing (Fully Furnished), House Size (sqm) (60), Water Supply (Borehole), Lot Size (sqm) (400), PHCN Availability (Poor), and Year Built (2010). A 'Predict Manual Rent' button is at the bottom left. A green banner at the bottom displays 'Estimated Annual Rent: ₦599,382' and an 'Activate Windows' watermark.

Manual Prediction Output

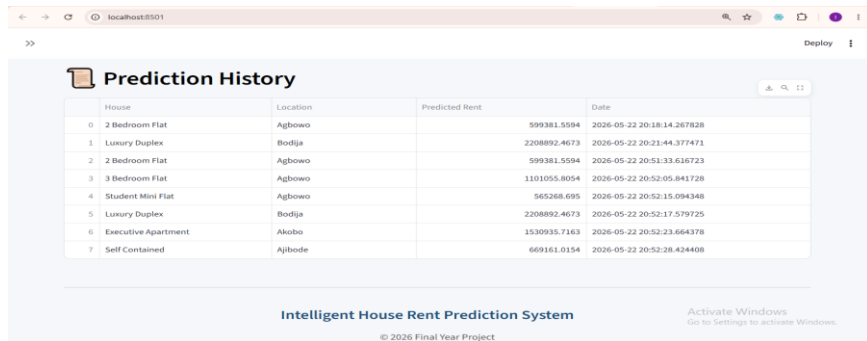


A web browser window showing the 'Compare Houses' interface. It has a navigation bar with links: Browse Properties, Manual Prediction, Market Analytics, Prediction History, Compare Houses (active), and About System. The main area has two dropdowns: 'Select First House' (Executive Apartment) and 'Select Second House' (Luxury Duplex). A 'Compare Houses' button is below. A table compares features for the two properties.

	Feature	Executive Apartment	Luxury Duplex
0	Location	Akobo	Bodija
1	Bedroom(s)	3	4
2	Bathroom(s)	2	3
3	Predicted Rent	₦1,530,936	₦2,208,892

An 'Activate Windows' watermark is visible at the bottom right.

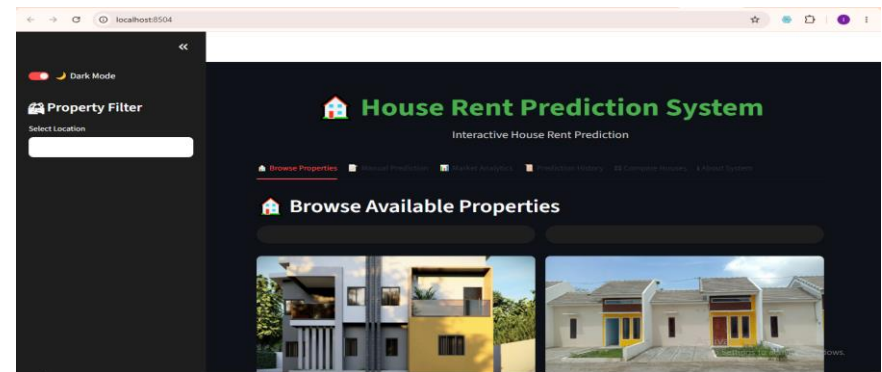
House Comparison Interface



A web browser window showing the 'Prediction History' interface. It displays a table of past predictions with columns for House, Location, Predicted Rent, and Date. The footer includes 'Intelligent House Rent Prediction System' and '© 2026 Final Year Project'.

House	Location	Predicted Rent	Date
0 2 Bedroom Flat	Agbowo	599381.5594	2026-05-22 20:18:14.267828
1 Luxury Duplex	Bodija	2208892.4673	2026-05-22 20:21:44.377471
2 2 Bedroom Flat	Agbowo	599381.5594	2026-05-22 20:51:33.616723
3 3 Bedroom Flat	Agbowo	1101055.8054	2026-05-22 20:52:05.841728
4 Student Mini Flat	Agbowo	565268.695	2026-05-22 20:52:15.094348
5 Luxury Duplex	Bodija	2208892.4673	2026-05-22 20:52:17.579725
6 Executive Apartment	Akobo	1530935.7163	2026-05-22 20:52:23.664378
7 Self Contained	Ajilbode	669161.0154	2026-05-22 20:52:28.424408

Prediction History Interface



Dark Mode Interface

Conclusion

In conclusion the data was gotten mainly for Ibadan and preprocessing was done, the model was built using linear regression, K-fold was used and there was model evaluation and an interactive streamlit user interface.

References

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THANK YOU